RNAfold Deck

Sidebar page **RNAfold Deck**: RSAS Excel results → hairpins of the chosen tier(s) → full-length and/or hairpin structures at 1–5 temperatures → a PowerPoint with one verified slide per fold.

Integrated in RSAS 4.0 from RNAfold Deck 1.3.0. The webserver fold and the Chrome screenshot were replaced by a local engine, so **no internet connection or browser is needed**.

What changed versus RNAfold Deck 1.3.0

Step	1.3.0	RSAS module
Fold	POST to rna.tbi.univie.ac.at, poll the job page	`RNA.fold_compound` with the webserver's model: `-p -d2 --noLP`, Turner 2004
Drawing	download the webserver SVG, screenshot it in headless Chrome/Edge	naview layout + base-pair-probability colors drawn locally (SVG + 300 dpi PNG, matplotlib)
Time	~15–25 min per 100 hairpins	~1–1.5 s per fold, mostly drawing
Checks	page/drawing sequence and temperature	the same checks on the local fold, plus MFE/structure cross-check against the RSAS sheet

The local result is the webserver's result: `tests/test_deck_local.py` re-folds five saved webserver jobs and gets the same MFE, dot-bracket, per-nucleotide colors and naview coordinates.

Use

1. Run an analysis on **Analyze** (writes `Data/Outputs/rna_results.xlsx`), or export results as Excel.
2. **RNAfold Deck** page → **Use latest RSAS results** (the last analysis, `~/RSAS/<input>_<ts>/<input>_rna_results.xlsx`; the deck is then saved inside that analysis folder as `<input>_RNAfold_deck_<ts>/`), or **Add Results file...** for exported workbooks, one per round in slide order.
3. Tick tiers, choose Full-length / Hairpin / both, fill Temp 1–5, press **Run**.

Command line (same engine):

```
python -m RnaThermofinder.deck run --results "~/RSAS/<input>_<ts>/<input>_rna_results.xlsx" --round-name R1 \
    --tier "Tier 1" --fold both --temp 25 --temp 37 --temp 42 --out-dir ~/Documents --name "RNAfold deck"
```

Output (`<output folder>/<deck name>/`)

File	Content
`<name>.pptx`	title slide per round, one slide per hairpin × fold × temperature
`<name> Tier 1 selection.xlsx`	selected Hairpin Analysis rows and their Full Sequence rows
`<name> verification.csv`	per slide: sequence, MFE, ensemble energy, structure, RSAS cross-check, all checks
`<name> structures/`	per fold: `<input>_NNN_<id>[_hp][_<T>C]` `.json` (fold data), `.svg` (vector, editable), `.png` (slide image)
`<name> log.txt`, `<name> mapping.json`	run log, traced IDs

`mfe_same_as_rsas` / `structure_same_as_rsas` compare with the RSAS sheet at the same temperature. For hairpin folds RSAS drops the trailing unpaired 3' tail before folding, so that comparison folds the trimmed hairpin too; the slide shows the hairpin exactly as in `Hairpin_Sequence`.

Files

```
RnaThermofinder/deck/
  local_fold.py   ViennaRNA fold, probability colors, naview layout, SVG/PNG drawing
  engine.py       select → trace → fold → build + verify (from RNAfold Deck 1.3.0)
  panel.py        DeckPanel (RSAS page) and standalone window: python -m
RnaThermofinder.deck.panel
  cli.py          python -m RnaThermofinder.deck ...
tests/test_deck_local.py, tests/deck_webserver_fixtures/
```